

Chapter 2

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Comparing Competing Theories in GrammarXiv

A case study on HPSG and categorial grammar

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Abstract: Comparing different theoretical frameworks is often a formidably complex—if not totally impossible—task. The goal of this chapter is to explore the possibilities of GrammarXiv (grammarxiv.net; an open-access database of linguistic ‘truths’) for this purpose, taking the comparison of HPSG and categorial grammar (Kubota, 2021) as a test case. The paper illustrates how the graph-based knowledge representation native to GrammarXiv enables an intuitive visualization of highly complex relations between linguistic phenomena, hypotheses and publications within and across competing theories, thereby offering an attractive alternative to the traditional prose-based exposition adopted in Kubota (2021). An additional advantage comes from the open-access, web-based platform of GrammarXiv, which offers a quick and flexible search interface and user-friendly visualization of search results that make domain-specific complex knowledge highly accessible for novices and outsiders.

Keywords: GrammarXiv, graph database, cypher, philosophy of linguistics, generative syntax

PACS: ...

1 Introduction

GrammarXiv (grammarxiv.net) is an open-access database, currently in development, which aims to collect linguistic ‘truths’—ideas and findings in the academic community of linguistics. The developing team is now working on making it available for open use in the community soon (within a year or so).

In linguistics (as in any other active field of academic research), the way research is done is primarily through debates over specific issues involving a complex network of empirical phenomena, generalizations, hypotheses, and theoretical frameworks which jointly constitute individual researchers' (scholarly) beliefs. As a result, understanding the landscape of research can be challenging not only for scholars outside the field but even for experts. GrammarXiv aims to address this issue by utilizing a graph database. Compared to traditional data structures like relational databases, graph databases allow for intuitive visualization of complex relationships between entities and concepts. Compared to the traditional approach to accumulating/summarizing scholarly knowledge such as overview articles, handbooks and survey monographs, an open-access, web-based platform has the advantage of rapid update, and easy and flexible search capabilities. Grammarxiv is still in its infancy, but its developers believe that it has a potential of changing the way research is done in linguistics.

This chapter conducts a case study examining whether GrammarXiv can achieve one of its main goals: visualizing complex knowledge in linguistics (the other goal is to offer an SNS platform, a function that is still at an early stage of development). For this purpose, I take the comparison of HPSG and categorial grammar as an example. The two theories are both non-mainstream theories of syntax, and a somewhat uncomfortable (and undeniable) reality in the field is that, while these theories offer interesting insights to the wider linguistic community, they tend to be perceived as something that belongs to the category 'for connoisseurs only', that is, something which, if you belong to the majority, you just don't need to care about. This is an unfortunate situation for everyone in the community (including both mainstream and non-mainstream syntacticians, as well as non-syntacticians), but it (ironically) offers an ideal subject matter for our present case study.

2 Overview of GrammarXiv

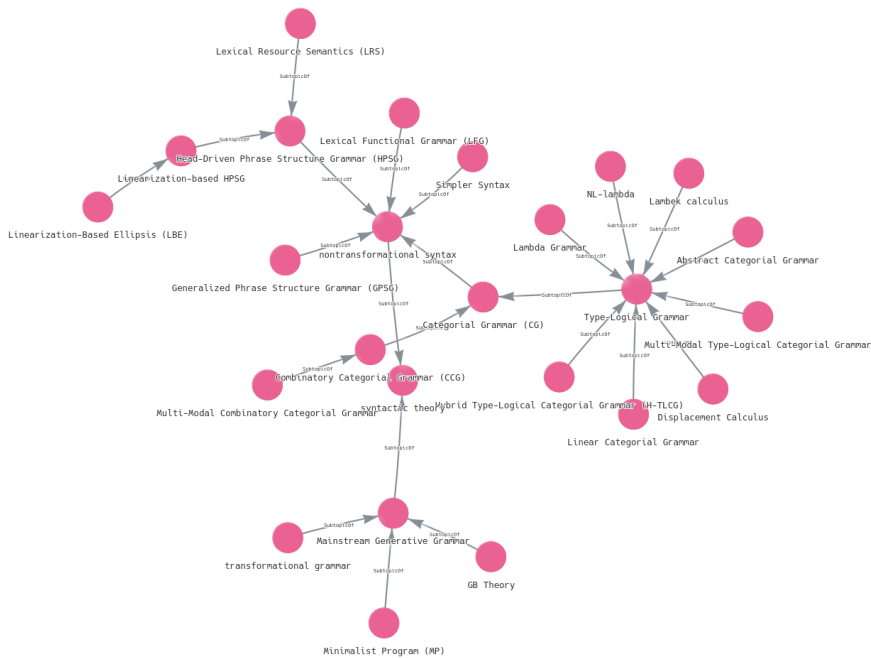
GrammarXiv is a database of academic knowledge in linguistics. It aims to systematically structure the knowledge of experts in the field, thereby enabling intuitive visualization and flexible search. The motivation for this project is to facilitate constructive criticism and interdisciplinary collaboration by organizing domain-specific knowledge in a way that is easy to access not only for specialists but also for non-specialists. To achieve this goal, Grammk adopts a graph database at its core. The key idea behind

this choice is that human knowledge is not stored in a uniform, neatly organized format like relational databases but is structured via relationships between entities, concepts, and statements or claims about these entities and concepts.

2.1 Some simple examples

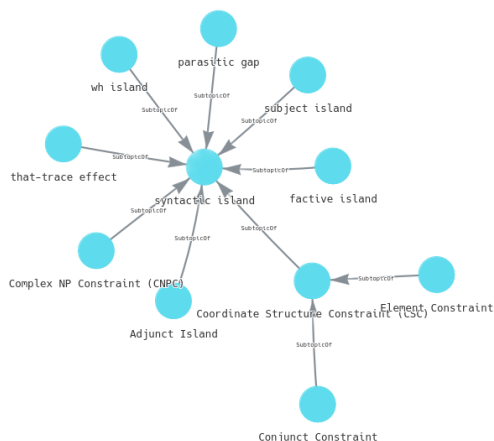
Representing knowledge via a network of relationships is a common method used to manage complex problems or phenomena in everyday situations. It makes such problems intuitively easier to understand and handle. Linguists should immediately see the benefit of this approach in some of the run-of-the-mill research scenarios like those illustrated in (1). In these examples, the search queries are written in the cypher language, the graph query language for the graph database called Neo4j (neo4j.com) adopted in GrammarXiv, and the reader can access the actual search results by clicking on the URL links provided.

- (1) a. **Show ‘parent-child’ relationships among different theoretical frameworks** (example: ‘syntactic theory’): [\[URL link\]](#)



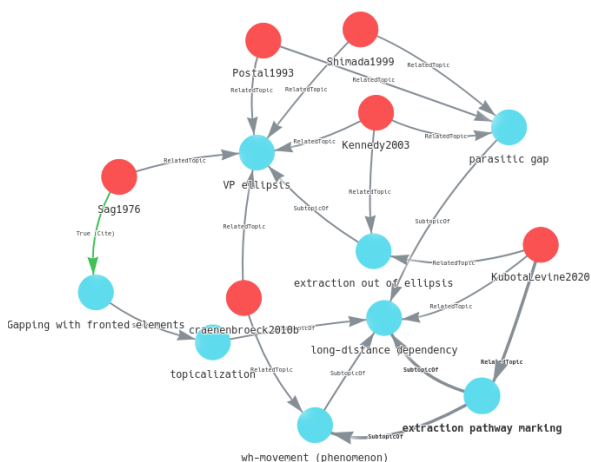
```
MATCH p = (:framework)-[:subtopic_of*]->({name: "syntactic theory"})
RETURN *, relationships(p)
```

- b. **Organize hierarchical relationships between empirical phenomena** (example: Ross's (1967) 'syntactic island') [\[URL link\]](#)



```
MATCH p = (:topic)-[:subtopic_of*]->({name: "syntactic island"})
RETURN *, relationships(p)
```

- c. **Retrieve publications that treat multiple phenomena in common** (example: 'VP ellipsis' and 'long-distance dependency') [\[URL link\]](#)

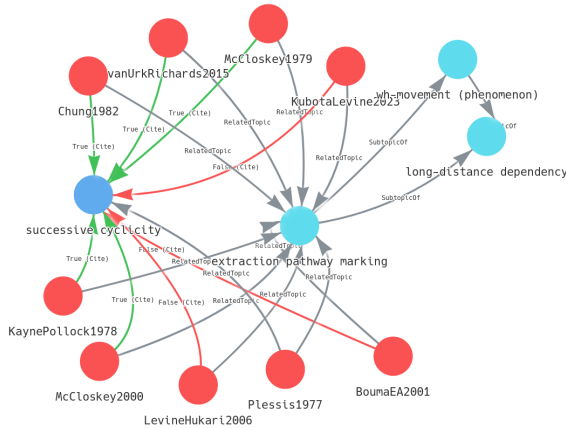


```

MATCH p = ({name: "VP ellipsis"}) <-[:subtopic_of*0..]- ()
-- (:publication) -- () -[:subtopic_of*0..]->
({name: "long-distance dependency"})
RETURN *, relationships(p)

```

- d. **Show relationships between phenomena, hypotheses and literature** (example: publications dealing with the phenomenon ‘long-distance dependencies’ and the hypothesis ‘successive cyclicity’) [\[URL link\]](#)



```

MATCH p = ({name: "successive cyclicity"})
<-[:subtopic_of*0..]- () -- (:publication) -- ()
-[:subtopic_of*0..]-> ({name: "long-distance dependency"})
RETURN *, relationships(p)

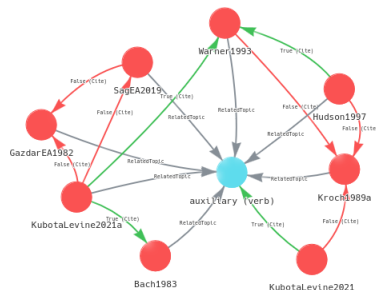
```

The simple illustration above should already make it clear that this type of relationship visualization is a very powerful tool. The guiding intuition here is that interesting and important ideas (to learn about, or to pursue further) can often be represented as *relationships* among what one might call ‘basic conceptual units’ (of various sorts and with varying degrees of abstractness) of academic knowledge, such as publications, hypotheses, phenomena, and examples/data.

But we can go further, by taking into account the nature of the relationships themselves. In a graph database, this can be done by subclassifying the relationships. Among such subcategories of relationships, the pair *truevote/falsevote* plays a central role in GrammarXiv.

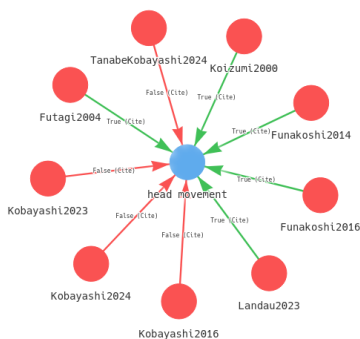
Truevote and falsevote are GrammarXiv jargons that are invented for the purpose of compactly expressing researchers' belief systems as complex networks of hypotheses (or assumptions). In most cases, 'truevote' and 'falsevote' can simply be read as 'support' and 'reject' (with slight metonymic extensions in some cases; see below for examples). Here are some examples:

- (2) a. **Truevotes/falsevotes among publications dealing with the same phenomenon** (example: publications on 'auxiliary (verb)') [\[URL link\]](#)



```
MATCH p = (a:publication) -- ({name: "auxiliary (verb)"})
-- (b:publication) -[:true|false]- (a)
RETURN *, relationships(p)
```

- b. **Truevotes/falsevotes among publications on a particular hypothesis** (example: controversies on 'head movement') [\[URL link\]](#)



```
MATCH p = (a:publication) -[:true|false]-
({name: "head movement"}) -[:true|false]- (b:publication)
RETURN *, relationships(p)
```

In examples like (2b) in which A is a publication and B is a hypothesis, ‘A truevotes/falsevotes B’ can be literally read as ‘A supports/rejects B’. When both A and B are publications, as in (2a), then the intended paraphrase is that A supports/rejects *a claim made in* B (note the subtleties involved in the use of existential quantification here; this entails that A can both truevote and falsevote B without contradiction). In a more extended sense, both A and B can be concepts, such as hypotheses. In such cases, the intended paraphrase is that whoever/whatever truevotes A *necessarily* truevotes/falsevotes B.

In many cases, all we need to do is just ask the database whether two entries are linked with each other. And if such a query isn’t specific enough, we can restrict the search by specifying the type of relation that holds between the entries in question. This should make clear the real power and flexibility of graph databases—essentially, once one designs the underlying representation scheme carefully, the data structure itself transparently embodies much of the abstract organization of knowledge. Thus, combining the search strategies illustrated above in (1) and (2) allows us to do something even more complex: it can guide novice users (without much domain-specific knowledge) to gradually internalize expert knowledge by simply *interacting* with the database.

Here is one example in which this potential of GrammarXiv can be exploited. In theoretical linguistics (as in any other field of empirical science), research activities typically center around debates among researchers over better explanations on observable phenomena, where competing explanations each consist of a network of hypotheses. Thus, for any given phenomenon, we can first ask (i) if any publication is related to it, and then, further ask about (ii) what hypotheses each publication argues for/against (truevote/falsevote). A cypher query can consist of multiple MATCH conditions (separated by comma). This functionality is useful when one wants to visualize different types of relations among multiple nodes simultaneously, as in this example.

(3) Debates on VP ellipsis [\[URL link\]](#)

MATCH

```
p = (a:publication) -- ({name: "VP ellipsis"}) -- (b:publication),
```

```
q = (a:publication) -[:true|false]-> (:hypothesis)
```

```
<-[:true|false]- (b:publication)
```

```
RETURN *, relationships(p), relationships(q)
```

The returned graph displays publications dealing with VP ellipsis and its subtopics and which are opposed to each other with respect to some hypothesis.

Essentially, GrammarXiv has the capability of allowing users to quickly generate customized visual maps of the research situation on any given topic. This should prove useful both for experts and novices, complementing the functionality of more traditional media for summarizing and organizing domain-specific knowledge such as handbook chapters and overview articles. In order for such a system to be useful, it has to be reasonably comprehensive in coverage and the way entries are registered in the database has to conform to some standard that ensures consistency and robustness. This is not an easy goal to achieve and GrammarXiv is still at an early stage of development, but we believe this to be fulfilled through a joint effort of the community once the system is made open to the public. We also envision the potential use of GrammarXiv to be much wider than linguistics, encompassing many disciplines in natural sciences and social sciences. One of the important future goals of the project is to extend its use case to the wider academic community.

2.2 Basic entries and relations in GrammarXiv

Before delving into the main discussion, I briefly summarize the basic types of entries and relations in GrammarXiv. Grammarxiv (version 1) posits the following seven basic entry types (here I ignore the extended component for the experimental SNS functionality):

- (4) a. **data** (subtypes: **acceptability**, **generalization**, **report**, **other**)
data entries correspond to claims about primary data. **acceptability** entries are example sentences with acceptability judgments (most of which are taken from published sources); **generalizations** are sets of such acceptability judgments (typically, multiple minimal pairs) exemplifying linguistic generalizations; **reports** are reports of empirical findings of various sorts (e.g., field work reports and meta-analysis reports).
- b. **publication** (subtypes: **article**, **book**, **incollection**, **inproceedings**, etc., following the BibTeX specifications)
publication entries contain bibliographic information of (published or unpublished) written scholarly work. Main properties specified for each subtype (such as **author**, **title**, **year**, **published_in**) largely follow the BibTeX specifications.
- c. **hypothesis**
This is for specific linguistic hypotheses of the usual sort.

d. framework

The framework type is used for identifying theoretical frameworks. frameworks usually consist of a set of mutually consistent hypotheses from which empirically testable implications are drawn on phenomena in a certain domain and which usually constitute belief systems shared by multiple researchers.

e. phenomenon (subtypes: keyword, language, vocabulary, research_question)

keyword entries correspond to specific linguistic phenomena; language entries are names of target languages; vocabulary entries name specific linguistic expressions; research_question identify both broad and specific research questions.

f. experiment (subtypes: judgment, comprehension, production, neural, other)

The experiment entries are a relatively new addition. They are used for identifying specific experiments reported in publications, with subtypes corresponding to different types of experiments.

g. author

Author information of publications is treated as distinct entries. The main reason for this design choice is that it enables writing search queries involving specific authors as key elements (e.g., ‘retrieve all the publications that argue against some publication written by author A’). GrammarXiv author IDs correspond to Semantic Scholar IDs.

In addition, the following relations between entries are posited:

- (5)
- a. truevote [P->DEFHP]
 - b. falsevote [P->DEFHP]
 - c. uncertainvote [P->DEFHP]: similar to truevote/falsevote, but neutral about the polarity of approval
 - d. can_explain [DEFHP->DEFHP]: this is when, e.g., a hypothesis can explain a piece of data
 - e. related_topic [X->T]: a versatile relation for just saying that two entries are related to each other in one way or another
 - f. entail [DEFH->DEFH]: this is the strict entailment relation such that ‘A entails B’ means B’s truth is *guaranteed* by A’s truth
 - g. refer_to [P->DEFHP]: this is most often the citation relation between publications, but the second argument can be of other types

- h. incompatible [DEFHP->DEFHP]: this means A entails not B and B entails not A
- i. subtopic_of [T->T]
- j. author_of [A->P]
- k. equivalent [X->X]: abbreviation for ‘A entails B and B entails A’

Here, the annotations such as [P->DEFHP] indicate which type of entry can stand in which argument position of the relation, with the following abbreviations: A: author, D: data, E: experiment, F: framework, H: hypothesis, P: publication, T: topic. For example, in the case of [P->DEFHP] for truevote, only publication entries can be the first argument and data, experiment, framework, hypothesis and publication entries can be the second argument.

3 Case study: Comparing HPSG and categorial grammar in GrammarXiv

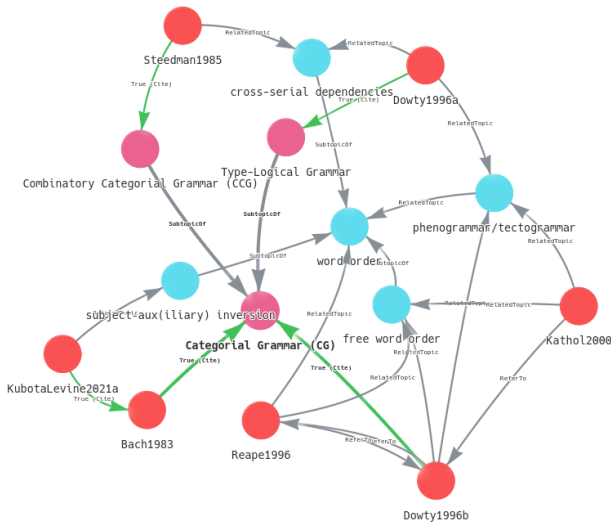
We can now address the central question of the present paper: can we profitably exploit the functionality offered by GrammarXiv for the comparison of different theories in real research situations? Our case study here will focus on a comparison between HPSG and categorial grammar. This choice is motivated by the following three reasons: (i) both theories are ‘non-mainstream’ syntactic theories and thus information about them is not easy to access; (ii) HPSG and categorial grammar are distinct theories but share many important properties, including emphasis on rigorous formalization and explicit syntax-semantics interface, making the relationship between the two not straightforward; (iii) a handbook article (Kubota, 2021) is available on the same topic, which makes it possible to compare GrammarXiv with the more traditional prose-based presentation. Three topics have been chosen from the coverage of Kubota (2021) for the illustration in the rest of this section: word order, long-distance dependencies and coordination. We explore the possibilities of the graph-based architecture and open-access platform embodied in GrammarXiv with respect to these topics and in relation to the handbook-style discussions in Kubota (2021).

3.1 Word order

The question of how to deal with languages with relatively free word order has been one of the major issues in contemporary syntactic theory. As discussed in Kubota (2021, Section 3.3.1), an influential idea on this issue in categorial grammar research is David Dowty's work in the 1980s (Dowty, 1982, 1996b), in which he proposed separating word order and syntactic structure by building on the notions of *phenogrammar* (surface word order) and *tectogrammar* (underlying valence-based combinatorics) due to Curry (1961). Dowty's work has directly influenced approaches to word order in later HPSG research in the 1990s.

Now, suppose we are interested in finding research on word order in HPSG and categorial grammar. In this kind of situation, writing queries such as the following is often useful:

- (6) Search for publication entries near 'Categorial Grammar (CG)' and 'word order' [\[URL link\]](#)



```
MATCH p = ({name: "Categorial Grammar (CG)"}
  -[*..2]- (:publication) -[*..2] ->({name: "word order"}))
RETURN *, relationships(p)
```

This query retrieves all publication entries that are within two hops from the two entries 'Categorial Grammar (CG)' and 'word order'. In what follows

we say that ‘Node A is *near* Node B’ if Node A is within a few hops from Node B.

This search result already identifies some work in HPSG, such as [Reape \(1996\)](#) and [Kathol \(2000\)](#), which are related to [Dowty’s \(1996a\)](#) work. From this, we can infer the influence that the latter has had on the former in the historical development of the two theories.

We can make the map more complete by explicitly adding work on word order in HPSG, that is, publication entries that are near ‘word order’ and ‘Head-Driven Phrase Structure Grammar (HPSG)’.

(7) **Research on word order in categorial grammar and HPSG**
[\[URL link\]](#)

```
MATCH p = ({name: "Categorial Grammar (CG)"}-[*..2]- (:publication)
           -[*..3] ->({name: "word order"}),
q = ({name: "Head-Driven Phrase Structure Grammar (HPSG)"}
     -[*..2]- (:publication)-[*..3] ->({name: "word order"}))
RETURN *, relationships(p), relationships(q)
```

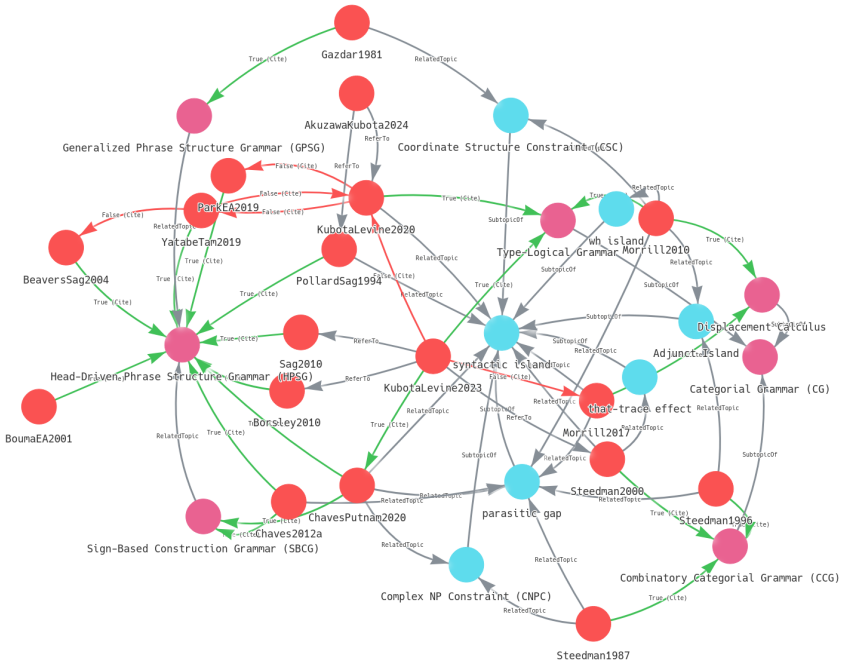
3.2 Long-distance dependencies

The analysis of long-distance dependency constructions, such as *wh*-movement in English, has been a central issue in contemporary syntactic theory. In particular, in relation to the issue of locality of syntactic movement in the mainstream syntactic literature, extensive research has been conducted on the status of so-called ‘island constraints’, and the notion of ‘successive cyclicity’. Long-distance dependencies constitute an important issue in HPSG and categorial grammar research as well. Specifically, these theories eschew the notion of ‘syntactic movement’, which means that standard account of these phenomena in the mainstream derivational approach is not available. This has basically led to two types of responses in each theory: (i) simulate the movement-based analysis in one way or another within non-movement theories; or (ii) question the underlying assumptions about the validity of the movement-based analyses of these phenomena (including reassessment of the empirical data themselves).

To get an overview of research on island constraints in HPSG and categorial grammar, we can adopt the same search strategy that has proven effective in our discussion of word order above, specifically, a ‘nearby’ search involving a theory name and a phenomenon name.

(8) **Work on syntactic islands in HPSG and categorial grammar**
[\[URL link\]](#)

```
MATCH p = ({name: "Categorial Grammar (CG)"}-[*..2]-(:publication)
-[*..2]->({name: "syntactic island"}),
q = ({name: "Head-Driven Phrase Structure Grammar (HPSG)"}
-[*..2]-(:publication)-[*..2]->({name: "syntactic island"}))
RETURN *, relationships(p), relationships(q)
```



This query retrieves the key literature on island constraints in the literature of HPSG (including its precursor GPSG) and categorial grammar such as Gazdar (1981); Steedman (1996, 2000); Morrill (2010); Pollard and Sag (1994); Sag (2010); Chaves (2012); Kubota and Levine (2020). The graph also shows which subtypes of island constraints (*that*-trace effect, Adjunct Island, Coordinate Structure Constraint, etc.) each publication deals with. Kubota (2021) does not discuss these subtypes due to space limitations. Thus, the ability to display such detailed information at a glance is an advantage of GrammarXiv.

In syntactic theory, island constraints have traditionally been regarded as syntactic constraints. However, this view has come under increasing scrutiny

over the years, with various researchers proposing alternative explanations of specific subtypes of constraints based on extragrammatical factors such as pragmatic felicity and on-line sentence processing load (see, e.g., [Newmeyer \(2016\)](#); [Chaves \(2021\)](#) for overviews). The treatment of island constraints in HPSG and categorial grammar is no exception of this history and recent developments in syntactic theory. In early research, it was common to account for island constraints within the grammar, but more recently researchers have increasingly sought explanations in terms of pragmatics or processing factors, making the combinatoric component of grammar simpler.

GrammarXiv posits two hypothesis entries ‘syntactic account of islands’ (the traditional view) and ‘nonsyntactic account of islands’ (the newer, extragrammatical view on islands). Approaches to syntactic islands in HPSG and categorial grammar advocating these two different perspectives can be retrieved by the following queries involving these hypothesis names as key elements:

- (9) a. **syntactic account of islands in categorial grammar and HPSG** [\[URL link\]](#)

```
MATCH p = ({name: "Categorial Grammar (CG)"}-[*..2]-(:publication)
           -[*..2]->({name: "syntactic account of islands"}),
      q = ({name: "Head-Driven Phrase Structure Grammar (HPSG)"}
           -[*..2]-(:publication)
           -[*..2]->({name: "syntactic account of islands"}))
RETURN *, relationships(p), relationships(q)
```

- b. **nonsyntactic account of islands in categorial grammar and HPSG** [\[URL link\]](#)

```
MATCH p = ({name: "Categorial Grammar (CG)"}-[*..2]-(:publication)
           -[*..2]->({name: "nonsyntactic account of islands"}),
      q = ({name: "Head-Driven Phrase Structure Grammar (HPSG)"}
           -[*..2]-(:publication)
           -[*..2]->({name: "nonsyntactic account of islands"}))
RETURN *, relationships(p), relationships(q)
```

Another major topic pertaining to long-distance dependencies is the notion of cyclicity ([Chomsky, 1973](#)), that is, the idea that *wh*-movement takes place in a stepwise manner. In the mainstream syntax literature, the so-called extraction pathway marking phenomena (such as complementizer choice in Irish; [McCloskey 1979](#)) have been argued to provide evidence for this notion (see, e.g., [van Urk \(2020\)](#) and [den Dikken \(2017\)](#) for overviews from somewhat different perspectives). Nontransformational syntactic theories do not

adopt the view that *wh*-dependency in natural language involves the notion of ‘syntactic movement’, but extraction pathway marking is a theory-neutral empirical issue that is relevant for all syntactic theories.

As mentioned briefly in Kubota (2021), the treatment of extraction pathway marking was an extensively debated topic in HPSG at one (relatively late) point of the development of the theory, in relation to a major architectural choice of whether to treat adjuncts on a par with arguments as elements that are listed on the verb’s lexical valence list (Bouma et al., 2001; Levine and Hukari, 2006). The way different publications are related to each other in this controversy can be visualized via the following query:

- (10) **Who works on extraction pathway marking in HPSG? [URL link]**

```
MATCH p = ({name: "Head-Driven Phrase Structure Grammar (HPSG)"})
          -[*.2]-(:publication)
          -[*.2]->({name: "extraction pathway marking"})
RETURN *, relationships(p)
```

In contrast to this situation in HPSG, an in-depth analysis of extraction pathway marking has long been lacking in categorial grammar, as noted by Kubota (2021) in the following quote:

there is currently no explicit treatment of this phenomenon in the CG literature. ... Extraction pathway marking poses a much trickier challenge to TLOG, in which extraction is typically handled by a single-chain movement-like process by means of hypothetical reasoning ... (Kubota, 2021, 1359–60)

GrammarXiv can tell us whether this situation has changed since then:

- (11) **Who works on extraction pathway marking in categorial grammar? [URL link]**

```
MATCH p = ({name: "Categorial Grammar (CG)"})
          -[*.2]-(:publication)
          -[*.2]->({name: "extraction pathway marking"})
RETURN *, relationships(p)
```

This search identifies Kubota and Levine (2024), an unpublished manuscript (at the time of writing) which proposes an analysis of extraction pathway marking in Type-Logical Grammar. This shows that Kubota (2021), which was published only three years ago, is already partly outdated on some of its coverage. If the idea of open-access, community-driven development embodied in GrammarXiv functions as intended, its flexible search capabilities

should provide a powerful tool for researchers to keep up with the most recent developments in the field.

3.3 Coordination

Much research has been conducted on coordination in both HPSG and categorical grammar. To get a rough overview of this research, we can use the same type of query as in Sections 3.1 and 3.2.

(12) **Coordination in HPSG and categorical grammar** [\[URL link\]](#)

```
MATCH p = ({name: "Categorical Grammar (CG)"}
  -[*.2]-(:publication)-[*.2]-()
  -[:subtopic_of]->({name: "coordination"}),
  q = ({name: "Head-Driven Phrase Structure Grammar (HPSG)"}
  -[*.2]-(:publication)-[*.2]-()
  -[:subtopic_of]->({name: "coordination"}))
RETURN *, relationships(p), relationships(q)
```

This graph shows several subtypes of coordination, such as nonconstituent coordination and Gapping, and we can see that numerous publications are related to these specific subtypes.

We can then explore the subgraphs more closely, expanding relations that are not shown in the above graph. In particular, when one wants to take a closer look at specific phenomena, it is often useful to show the relations between publications and hypotheses. This will often help us identify (or at least guess) major controversies in the field with respect to the phenomenon in question. This can be done by the same type of search we performed at the end of Section 2. For example, the following query displays publications on Gapping and the major hypotheses they are related to:

(13) **Debates on Gapping** [\[URL link\]](#)

```
MATCH
  p = ({name: "Gapping"}) -- (a:publication)
  -[:true|false]-> (:hypothesis)
RETURN *, relationships(p), relationships(q)
```

We can modify this query slightly to also display the theoretical framework that each publication adopts. One way of doing this is to write a query that lists two paths, as follows:

- (14) **Debates on Gapping (showing related theoretical frameworks)** [\[URL link\]](#)

```
MATCH
  p = ({name: "Gapping"}) -- (a:publication)
                        -[:true|false]-> (:hypothesis),
  q = (a)-[:true]-()-[:subtopic_of*0..]->(e:framework)
RETURN *, relationships(p), relationships(q)
```

This query returns publications related to various theoretical frameworks, not just HPSG and categorial grammar. If we want to focus on the two theories only, we restrict the search further by adding an explicit constraint on the theory names:

- (15) **Debates on Gapping in HPSG and categorial grammar** [\[URL link\]](#)

```
MATCH
  p = ({name: "Gapping"}) -- (a:publication)
                        -[:true|false]-> (:hypothesis),
  q = (a)-[:true]-()-[:subtopic_of*0..]->(e:framework)
WHERE e.name = "Categorial Grammar (CG)"
      OR e.name = "Head-Driven Phrase Structure Grammar (HPSG)"
RETURN *, relationships(p), relationships(q)
```

When the database becomes large, simple queries such as (14) often return a large number of irrelevant relations and entries. In such situations, filtering the search results by imposing further conditions on the nodes in the query expression with the WHERE clause, as is done in (15), is often useful.

The following shows the same type of query for nonconstituent coordination, another subtype of coordination:

- (16) **Debates on nonconstituent coordination in HPSG and categorial grammar** [\[URL link\]](#)

```
MATCH
  p = ({name: "nonconstituent coordination (NCC)"})
      -- (a:publication) -[:true|false]-> (:hypothesis),
  q = (a)-[:true]-()-[:subtopic_of*0..]->(e:framework)
WHERE e.name = "Categorial Grammar (CG)"
      OR e.name = "Head-Driven Phrase Structure Grammar (HPSG)"
RETURN *, relationships(p), relationships(q)
```

4 Conclusion

This chapter examined how GrammarXiv, a graph-based open-access databaseplatform, can support comparison of research results across different theoretical frameworks. Using a case study of HPSG and categorial grammar, it was shown that GrammarXiv enables intuitive visualizations of complex network of scholarly knowledge, making it easier to explore connections among theories, hypotheses, and empirical phenomena.

Compared to traditional prose-based approaches (represented by handbooks and similar resources), GrammarXiv offers clear advantages in searchability, up-to-date coverage, and accessibility for both experts and novices. The case studies on three specific empirical domains—word order, long-distance dependencies, and coordination—illustrate how the structured representation of domain-specific knowledge in GrammarXiv facilitates focused investigation of theoretical debates.

Although still under development, GrammarXiv has the potential to reshape scholarly practice in linguistics and beyond. Its success will depend on community engagement and systematic data curation, but its design offers a promising foundation for more transparent and collaborative research.

Acknowledgment: ...

References

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